

Radiation Recall and Enhancement Reactions

RADIATION RECALL REACTION

Recall reactions present as erythema confined to an area of prior radiation exposure or sunburn. In severe cases, vesiculation may occur. True radiation recall can develop years after the initial irradiation, and its pathogenesis remains unknown. It is hypothesized that radiation-damaged keratinocytes are further injured by the administration of a cytotoxic agent.

Methotrexate frequently causes a unique recall reaction when administered 1 to 3 days after sun exposure, although cases have been reported as late as 8 days post-exposure. Folinic acid supplementation (leucovorin rescue) does not prevent methotrexate-induced UV sunburn recall reactions.

RADIATION ENHANCEMENT REACTIONS

Radiation enhancement refers to a synergistic interaction between an antineoplastic agent and radiation therapy. This radiosensitizing effect is often therapeutically desirable but can lead to adverse cutaneous effects. The result is typically radiation dermatitis, which begins with pain and erythema, followed by desquamation and hyperpigmentation.

Although several chemotherapeutic agents can cause this reaction, **doxorubicin** and **dactinomycin** are most commonly implicated.

Cytotoxic Agents with Radiation Reactions

- Photosensitivity
- Dacarbazine
 - Fluorouracil

- Hydroxyurea
- Methotrexate
- Mitomycin
- Thioguanine
- Vinblastine

- **Radiation enhancement**

- Bleomycin
- Cisplatin
- Dactinomycin
- Doxorubicin
- Fluorouracil
- Hydroxyurea
- Methotrexate

- **Radiation recall reaction**

- Bleomycin
- Cytarabine (common)
- Dactinomycin
- Doxorubicin
- Etoposide
- Fluorouracil
- Hydroxyurea
- Lomustine
- Methotrexate
- Melphalan
- Mitomycin
- Paclitaxel
- Sunburn recall reaction: Methotrexate, Suramin

Agents in bold represent systemic therapies frequently used by dermatologists in the management of cutaneous diseases.

▲ FIGURE 228-8 Reactivation of solar erythema 2 days after high-dose methotrexate administration in a young woman with choriocarcinoma.

Dermal Complications

LOCAL INJURY

Extravasation of chemotherapeutic agents may occur in up to 6% of adults receiving intravenous infusions and appears to be even more common in pediatric patients. The severity of injury depends on the specific drug, rate of delivery, concentration, and volume extravasated.

Irritants

Agents with minimal toxicity upon extravasation may cause only chemical cellulitis or phlebitis and are classified as irritants.

- **Chemical cellulitis** presents as erythema, induration, and pain at the injection site, sometimes tracking along the vein. It is typically self-limited and resolves without treatment.
- **Chemotherapy-induced phlebitis** manifests as tender linear cords along the vein.

Vesicants

Vesicants cause direct tissue necrosis when extravasation occurs and can lead to severe complications.

Management of Extravasation:

- Immediately discontinue the infusion.
- Apply either hot or cold packs depending on the agent:
- **Cold packs** are used for most chemotherapeutic agents to localize the drug and promote degradation.

- **Hot packs** are indicated for **vinca alkaloids**, as cold application increases the risk of tissue necrosis.